Zihan Lin

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EDUCATION

Georgia Institute of Technology Jan 2026 - May 2028 (Expected)
B.S. Biomedical Engineering *Atlanta, GA*

University of Florida Aug 2024 - Dec 2025
Biomedical Engineering | Dean's List *Gainesville, FL*

RESEARCH EXPERIENCE

Undergraduate Researcher | Robotic Berry Harvesting Jun 2026 - Present
BM2 Lab, Georgia Tech | PhD Mentors: Yixuan Xia, Yilin Cai *Atlanta, GA*

Simulation Platform Development & Integration | RRP Simulation & ROS 2 Integration

- Independently built the original harvesting simulation from chassis-mounted RRP arm CAD; authored URDF/MJCF models with joints, collisions, and gripper constraints; corrected frames and checked kinematics across 10 poses and 15 frames.
- Integrated ROS 2 Humble/ros2_control with MoveIt 2 for arm/gripper execution and joint feedback; configured OMPL/Pilz planning and verified initial NVIDIA cuMotion trajectories in MuJoCo.
- Created seed-reproducible scenes with 96 dynamic fruits, contact and attachment constraints, synchronized MoveIt collision objects, and scene-reset and single-reach interfaces.
- Published simulated ZED 2 and gripper-mounted D435 RGB-D streams over ROS 2; aligned CAD camera frames with ROS optical conventions for multi-view visualization in RViz.

Dual-Arm Integration | OpenArm, Compliant Contact & Force-Control Research

- Extended the original six-link gripper to OpenArm dual arms, building a 16-actuator rigid-body model with passive joints, closed-chain constraints, and shared URDF/MJCF mount offsets.
- Prototyped articulated belts, compliant stems, and load-/fatigue-based fruit detachment; preparing normal/tangential force, deformation, and tension logging for force-control studies and dual-arm integration.

Undergraduate Researcher | MRI-Compatible Needle Placement Jan 2026 - May 2026
BM2 Lab, Georgia Tech | PhD Mentor: Samuel Wilcox *Atlanta, GA*

- Reconstructed and redesigned a pneumatic actuator assembly in SolidWorks from hardware and STEP references for MRI-compatible needle placement.
- Refined actuator layouts, ball-joint linkages, and pivot/tilt mechanisms through CAD iterations and design reviews to address needle-angle and workspace constraints.

ROBOTICS LEADERSHIP & PROJECTS

Simulation Lead, Software Team | RoboRambler Club 2026 - Present
Georgia Tech | RoboMaster ARCC *Atlanta, GA*

- Lead Sentry2026 simulation integration and debugging for RoboMaster ARCC; adding virtual cameras through URDF/Xacro and MJCF position/orientation configuration.
- Restored local ROS 2 startup and robot-description delivery by adjusting socket-buffer limits and adding simulation-only Cyclone DDS message/fragment settings.
- Instrumented C++ startup code to isolate a teammate's launch stall to GLFW adapter construction; scripted pinned-source patching and Pixi/colcon builds.
- Planned next scope: physics and coordination simulation for a dual-arm Engineer robot, plus a new double-yaw Sentry model.

Independent 6-DOF Robotic Arm | STM32, MuJoCo, Python Jun 2026 - Aug 2026
Independent Engineering Project *Shanghai, China*

- Designed and fabricated a 6-DOF desktop arm with custom 3D-printed joints, belt reductions, bearing-supported shafts, and a 24 V stepper/servo actuation system.
- Implemented STM32 control with dual absolute magnetic encoders, homing, calibration, and coordinated commands; resolved encoder wraparound, decoding, and UART timeout faults.
- Built Python telemetry and a MuJoCo arm model with damped-least-squares IK, multi-start search, and joint limits; generated Cartesian handwriting trajectories in simulation.

OTHER EXPERIENCE

Teaching Assistant, General Chemistry II | University of Florida Jan 2025 - May 2025

- Led weekly quantitative problem-solving sessions and office hours for 30+ students.

TECHNICAL SKILLS

Programming: Python, C, Java, MATLAB; C++ source-level debugging (basic)
Robotics & Simulation: ROS 2 Humble, ros2_control, MuJoCo, MoveIt 2, cuMotion, URDF/Xacro/MJCF, RViz, TF
Systems & Build: Linux/ROS 2 debugging, Cyclone DDS, Pixi, Git/GitHub, Docker, CMake, colcon
Hardware & CAD: STM32, UART, encoders, SolidWorks, Onshape, 3D printing, ZED 2, RealSense D435